CrohnIPI Diffusion: Class-Balanced Synthetic Augmentation for Crohn's Endoscopy Classification

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Motivation

Clinical endoscopy datasets are often highly imbalanced, where majority classes dominate the training data. This imbalance makes it difficult for classification models to learn rare Crohn's disease [1] findings such as ulceration, edema, and stenosis.

Our work investigates whether diffusion-generated synthetic images can help improve classification performance under severe class imbalance.

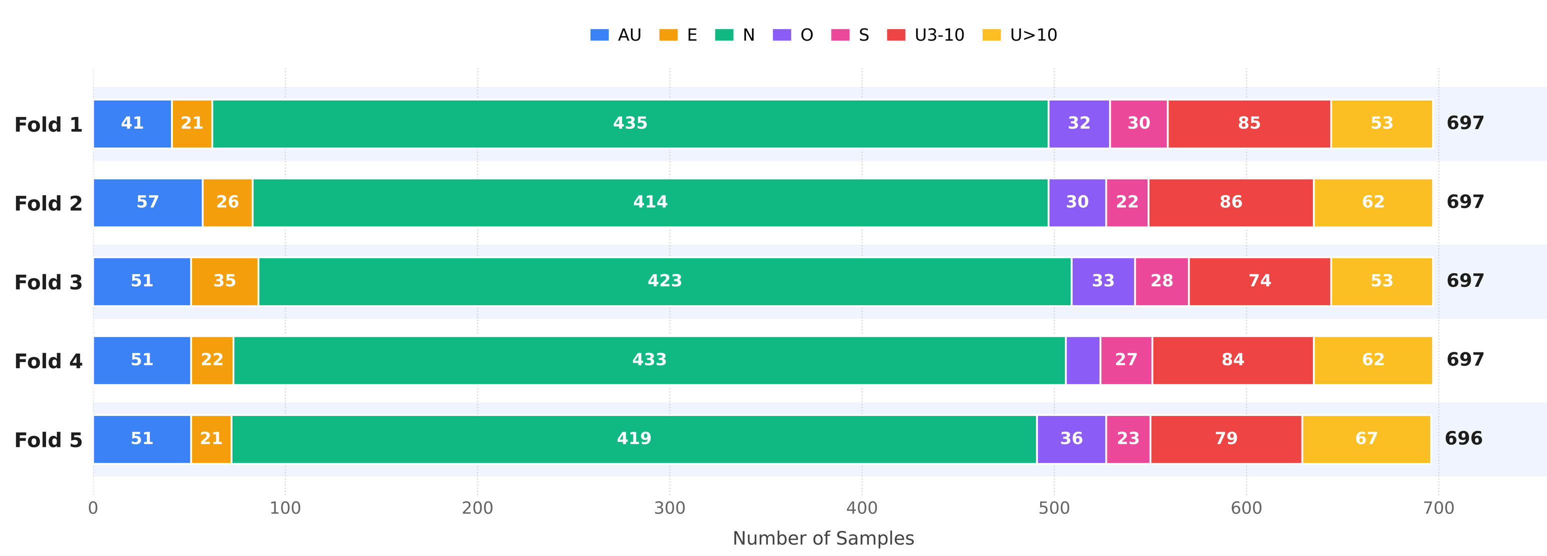

Figure 1. CrohnIPI dataset distribution across five folds. The normal class dominates the dataset, creating a strong class-imbalance problem.

Dataset

Dataset: CrohnIPI endoscopy image dataset

Classes:

Class	Full name
N	Normal
E	Erythema
O	Edema
AU	Aphthoid Ulceration
S	Stenosis
U3-10	Ulceration between 3 mm and 10 mm
U>10 / U_gt_10	Ulceration greater than 10 mm

Evaluation setup: 5-fold cross-validation

Main challenge: Severe class imbalance, where the majority class dominates and rare pathological classes have fewer examples.

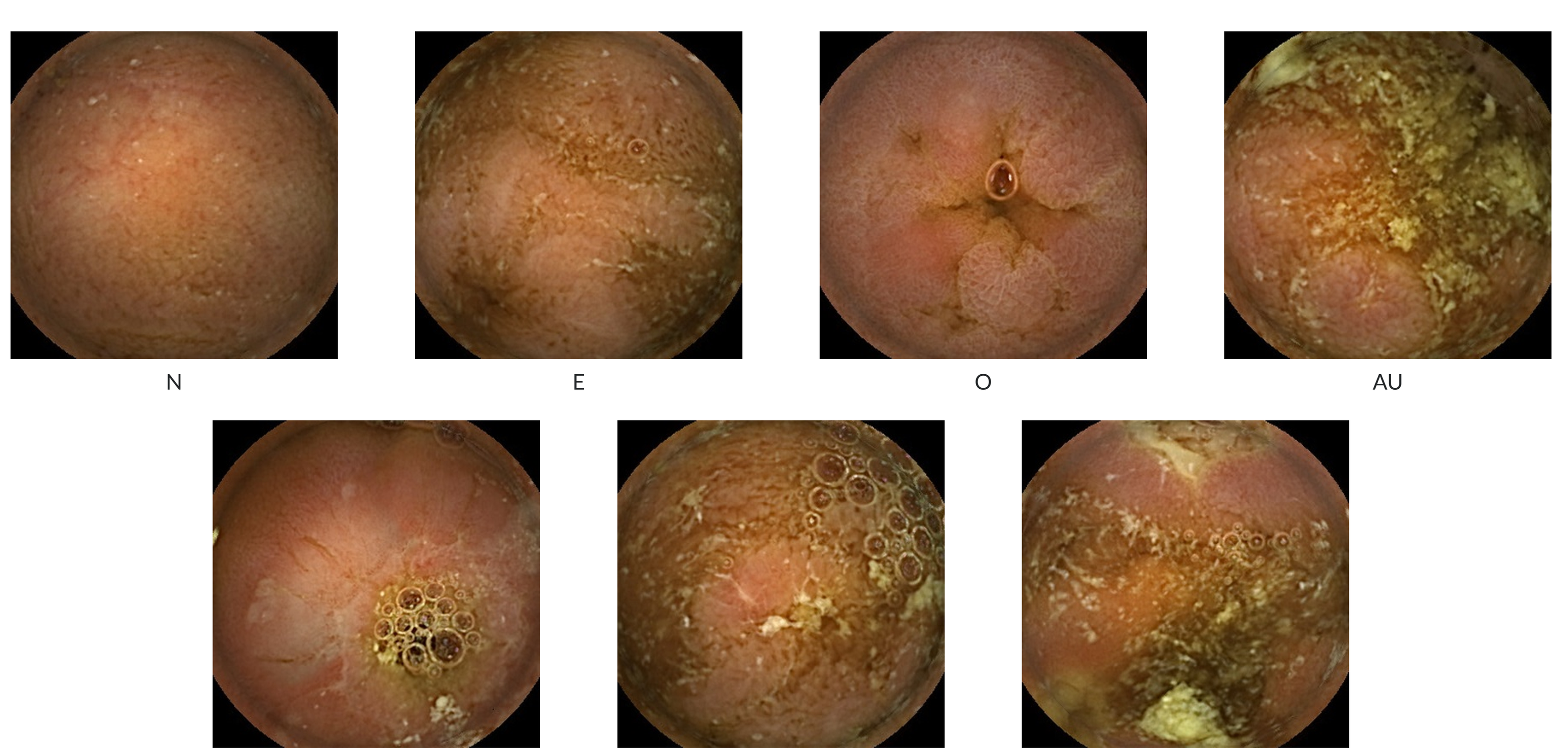

Figure 2. Representative CrohnIPI endoscopy examples from the seven dataset classes.

Research Question

Key question: To what extent can per-class diffusion-generated synthetic images improve the classification performance of CrohnIPI models under severe class imbalance?

Proposed Method

We train per-class LoRA diffusion [3] adapters using real CrohnIPI images. Each adapter generates synthetic images for a specific clinical class. The generated images are then passed through multiple filtering steps before being used to train a ConvNeXt-Large [2] classifier.

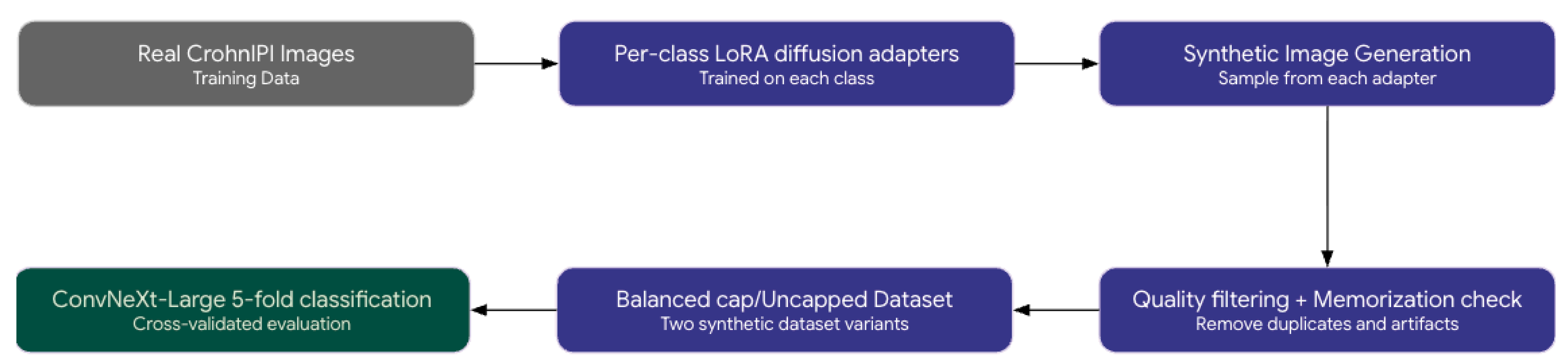

Figure 3. Overview of the proposed synthetic augmentation pipeline using per-class LoRA diffusion adapters and filtered synthetic data for classifier training.

Synthetic Image Filtering

To reduce low-quality or unsafe synthetic samples, we apply several filtering steps before using generated images for downstream classifier training.

Filtering Step	Purpose
1. Artifact detection	Remove blurry or broken-color images
2. Class-consistency check	Remove images predicted as the wrong class
3. DINOv2 memorization check	Detect synthetic images that are too similar to real images
4. Diversity deduplication	Remove highly similar synthetic samples
5. Final filtered set	Used for downstream classifier training

Table 1. Synthetic image filtering steps used to remove low-quality, inconsistent, memorized, or duplicate generated samples.

Filtering thresholds:

- Laplacian variance < 50
- Channel std < 0.02
- ConvNeXt-Tiny confidence < 0.70
- Pairwise cosine similarity >= 0.95

Experiment Design

We compare real-only training against standard augmentation, oversampling, and multiple synthetic augmentation settings.

Condition	Description
Real only	Baseline using only real images
Real + Augmented	Standard clinical image augmentation
Oversampling	Repeated rare-class real images
Synthetic uncapped	Full per-class LoRA synthetic dataset
Synthetic balanced/capped	Equal synthetic cap per class
Focal loss [4]	Synthetic data with focal loss [4]
Inverse-frequency CE	Synthetic data with class-weighted loss

Table 2. Training conditions used to evaluate whether synthetic diffusion augmentation improves CrohnIPI classification under severe class imbalance.

Synthetic Dataset Design

Synthetic images were generated for each class using per-class LoRA adapters. We evaluate both uncapped synthetic datasets and balanced/capped synthetic datasets to study whether synthetic class balancing improves classification performance.

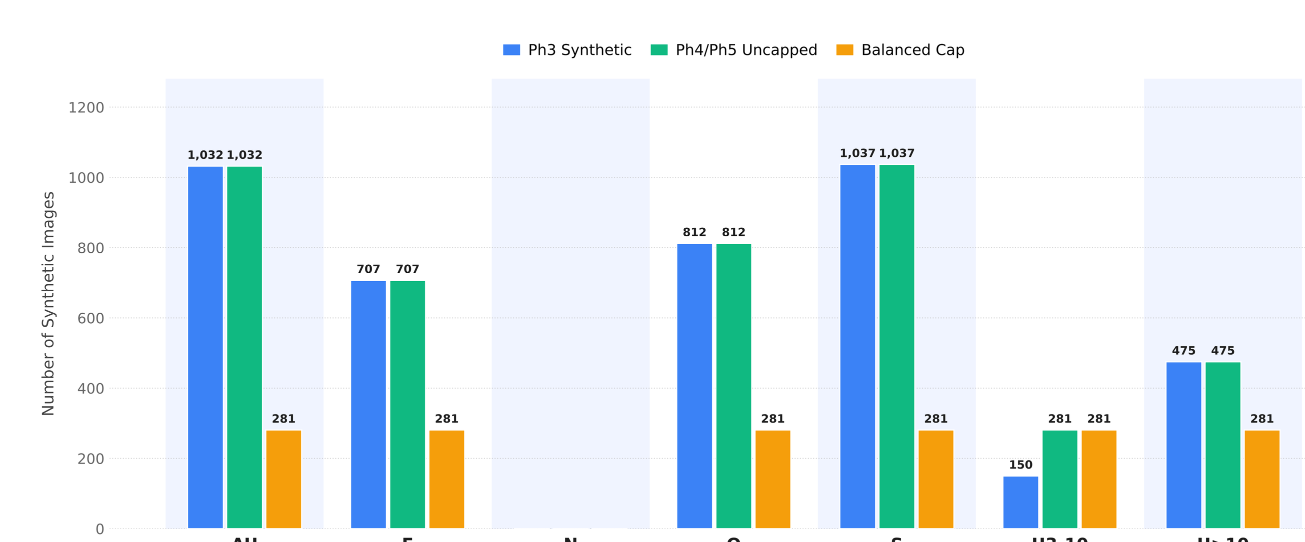

Figure 4. Number of generated synthetic images per class across different synthetic dataset variants.

Main Results

Method	Macro F1	Balanced Accuracy	Macro AUROC
Real only	0.7888	0.7781	0.9709
Real + Augmented	0.7973	0.7902	0.9727
Oversampling	0.7786	0.7596	0.9628
Synthetic	0.8085	0.7949	0.9708
Synthetic + Balanced	0.8076	**0.8090**	0.9591
Focal Uncapped	**0.8139**	0.8078	0.9708
Focal Balanced	0.8077	0.7969	**0.9721**

Table 3. Classification performance across real-only, augmentation, oversampling, and synthetic augmentation conditions. Focal Uncapped achieves the best Macro F1.

Key result: Synthetic augmentation improves Macro F1 over the real-only baseline. The best Macro F1 is achieved using the Focal Uncapped synthetic condition.

Per-Class Recall Analysis

Synthetic augmentation improves recall for several rare classes, but the gains are not uniform across all labels. This suggests that diffusion-based augmentation can help with class imbalance, but class-specific generation quality remains important.

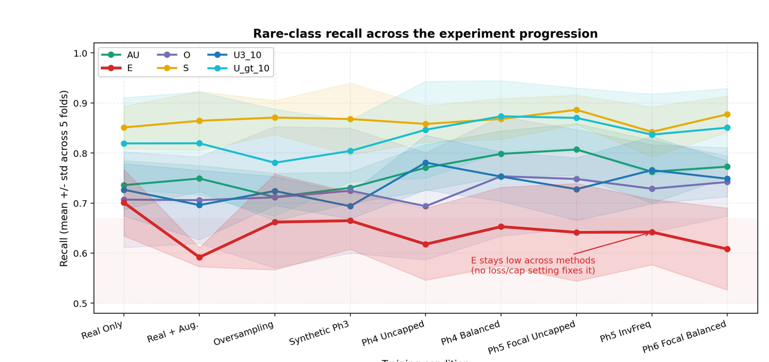

Figure 5a. Rare-class recall across training conditions. Synthetic augmentation improves some rare-class recalls, but not all classes benefit equally.

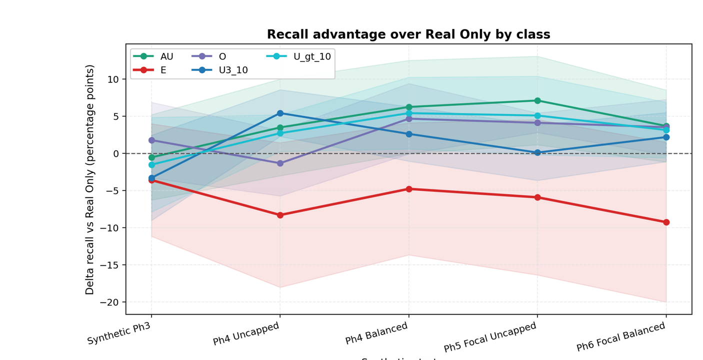

Figure 5b. Recall advantage compared with the real-only baseline.

Failure Analysis

The current method still struggles with the E class. Failure analysis shows that some E samples are confused with other classes, including the majority class. This indicates that synthetic data alone is not enough; better class-specific generation and expert-guided filtering may be needed.

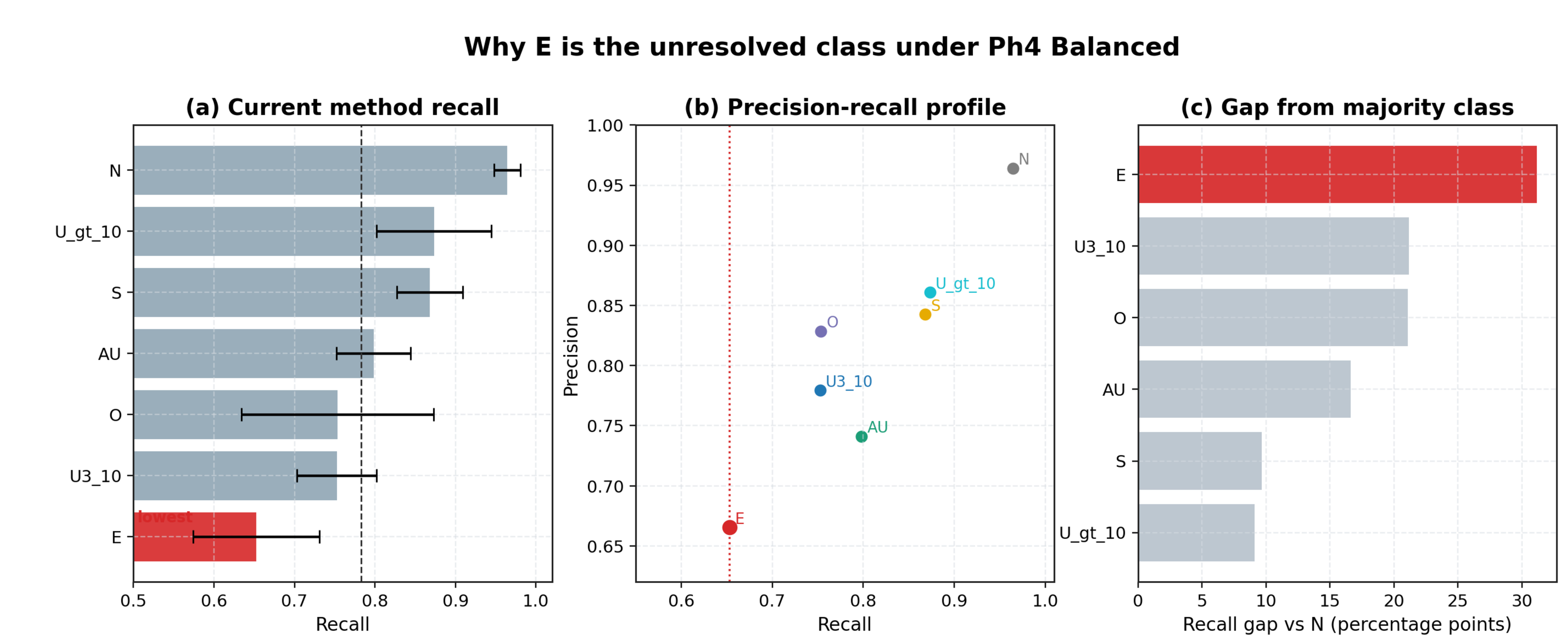

Figure 6. Failure analysis showing that the E class remains difficult under the current synthetic augmentation method.

Main Takeaways

- Per-class LoRA synthetic data improves class-imbalanced CrohnIPI classification.
- Synthetic augmentation outperforms oversampling; Focal Uncapped gives the best Macro F1.
- Balanced and focal settings improve complementary metrics.
- The E class remains challenging, motivating stronger filtering and expert validation.

References

[1] Vallée et al., CrohnIPI, SPIE Medical Imaging, 2020.

[2] Woo et al., ConvNeXt V2, CVPR, 2023.

[3] Podell et al., SDXL, ICLR, 2024.

[4] Mukhoti et al., Calibrating deep neural networks using focal loss, NeurIPS, 2020.